

Computer Vision for biomedical images: Amia public challenge 2026

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Abstract— This project investigates the use of deep learning for multi-label classification of fourteen thoracic abnormalities. An EfficientNet-based model was trained and different preprocessing and data augmentation techniques were evaluated. The final augmentation pipeline improved Macro PR-AUC by approximately 0.08 compared with the baseline model, although the magnitude of the improvement varied considerably across classes.



1 INTRODUCTION

1.1 Problem statement.

Chest X-ray images are widely used for examining the thoracic region and detecting abnormalities. However, the interpretation of chest radiographs can be challenging because of large variability in the appearance and location of abnormalities, as well as the presence of large anatomical structures such as the ribcage. This makes automated analysis of chest X-ray images a challenging computer vision problem.

The objective of this project is to automatically classify thoracic abnormalities in chest radiographs. The task is based on a provided dataset containing 15,000 chest X-ray images, of which 8,573 images are provided for training and 6,427 for testing. The training data contains annotations from radiologists identifying different types of thoracic abnormalities and their location by specifying a bounding box.

The main goal of this project is to develop and evaluate a deep learning approach capable of detecting these abnormalities from chest X-ray images. In total 14 different abnormalities are considered, and multiple abnormalities can occur in the same image. In addition to model performance, the project focuses on investigating how the characteristics of the dataset and different preprocessing, augmentation, and training choices affect the results.

1.2 Techniques used.

1.2.1 EfficientNet-B4 Backbone.

We selected EfficientNet[1] as the backbone of our classification model because of its strong image classification performance and efficient architecture. The EfficientNet family is based on compound scaling, which jointly scales the network's depth, width, and input resolution rather than increasing only one of these dimensions.

The model is initialized with weights pretrained on ImageNet, providing a useful starting point for learning visual features before adapting the network to chest

radiographs. This constitutes a transfer learning approach. Rather than training the network entirely from scratch, the pretrained representations are fine-tuned on our domain-specific dataset. In the project *timm*[2] is utilized to load the model.

1.2.2 Image preprocessing techniques.

Due to the highly imbalanced dataset, the low number of instances of some abnormalities and to improve the robustness of the model and avoid overfitting, different preprocessing strategies were investigated:

- Data augmentation
 - Scaling
 - Rotations
 - Contrast enhancement.
 - Gaussian noise
- Contrast enhancement and intensity preprocessing.
 - CLAHE
 - Gamma correction.
 - Percentile-based contrast normalization

2 METHODS

2.1 Dataset Analysis.

The provided dataset consists of 15,000 chest X-ray images, of which 8,573 images constitute the training set and the remaining 6,427 images form the test set. The images are in PNG format and have been resized to 1024×1024 pixels.

The dataset contains annotations for up to 14 different thoracic abnormalities. In addition to the images, three main CSV files are provided:

- **test.csv**: contains the identifiers of the images belonging to the test set.
- **train.csv**: contains the image identifier, the corresponding abnormality annotation, the bounding

box of the annotated abnormality, and the identifier of the radiologist who produced the annotation.

- **Img_size.csv:** contains the image identifier together with the original width and height of the image. This information is particularly important because the bounding box coordinates provided in train.csv are expressed in the coordinate system of the original image.

In train.csv, each row corresponds to an annotation made by a specific radiologist for a single abnormality in a given image. Each image was reviewed by three different radiologists. However, a single radiologist may identify multiple abnormalities in the same image, resulting in multiple annotations for the same image and radiologist. When a radiologist does not identify any abnormality, the annotation is recorded as "No Finding", while the bounding box fields are left empty.

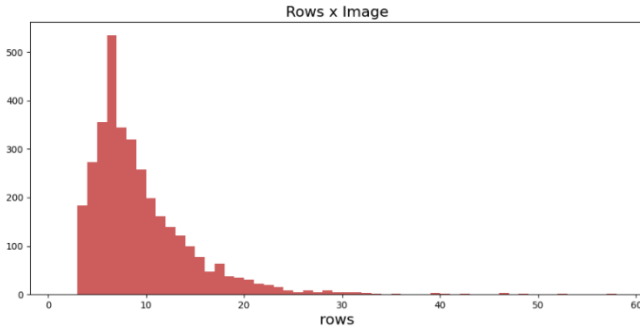


Figure 1: Number of images (y) that have (x) annotations.

2.1.1 Dataset quality.

One potential limitation of the annotations is the uncertainty associated with the radiologists' interpretations. The annotations are produced by different radiologists, there may be variability both in the identification of an abnormality and in the location assigned to it and make the learning task more challenging.

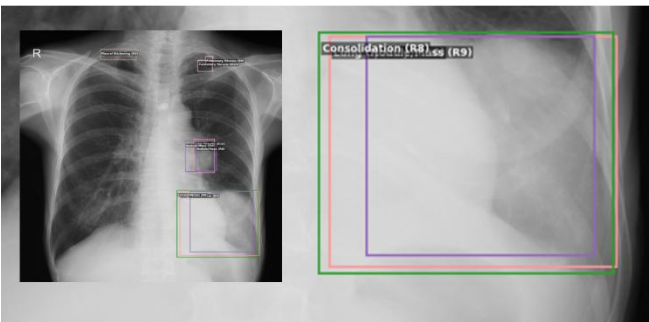
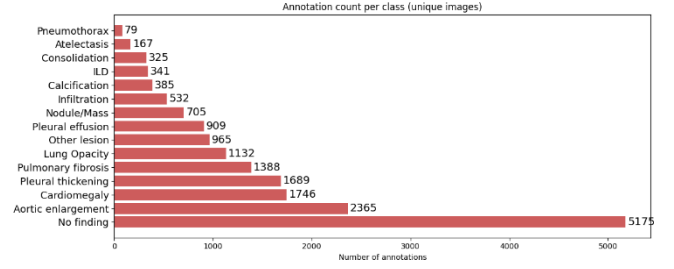


Figure 2: Example of a radiograph where the same abnormality has been labeled differently by the 3 radiologists.

The dataset is also strongly imbalanced across the different classes. Within the training set, some abnormalities, such as Pneumothorax and Atelectasis, are annotated in only 79 and 167 unique images, respectively. In contrast, more frequent classes such as Aortic enlargement and No Finding occur in 2,365 and 5,175 unique images, respectively. This substantial difference in class frequency can

bias the training process towards the more represented



classes and make the detection of rare abnormalities more difficult.

Figure 3: Number of unique radiographs where every class/abnormality appears annotated at least once.

Another characteristic that may affect model training is the variability in how the radiographs were acquired and processed. We observed that some images present differences in patient positioning, black and white borders or other visible artefacts or marks that introduce differences in image appearance that are unrelated to the presence of an abnormality.

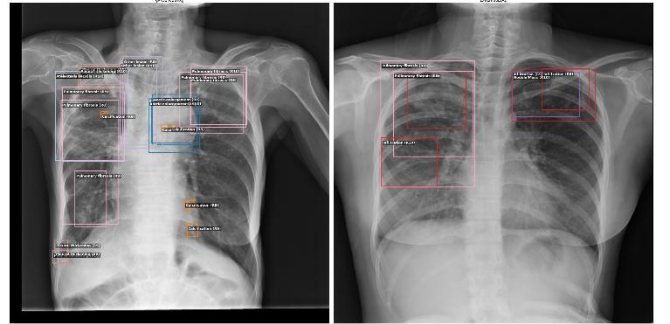


Figure 4: Example of a radiograph where multiple abnormalities have been identified and that present black borders to the left and bottom of the image.

2.2 Preprocessing of the dataset.

2.2.1 Cropping black and white image borders.

This preprocessing step was applied to reduce the influence of large visual regions that do not contain information related to the abnormalities being detected. Black and white borders can occupy a considerable portion of some images while providing no relevant information for the classification task. Their presence may encourage the model to learn acquisition or processing characteristics but features associated with thoracic abnormalities.

The cropping operation was performed before resizing. The preprocessing was applied once to all training and validation images, and the resulting cropped and resized images were stored in a separate directory for subsequent experiments. Applying them dynamically every time an image is loaded would require repeating the same preprocessing during every training epoch. In our experiments, performing these operations during data loading would have added several minutes to each training epoch.

2.2.2 Image resizing.

Due to hardware limitations during model training, the original 1024×1024 images were resized to 512×512

pixels. This reduction was necessary to make the training process computationally feasible with the available hardware. This results in a fourfold reduction in the total number of pixels and represents a potential loss of fine-grained visual information. This trade-off is relevant for this task because some of the abnormalities, such as *Nodule/Mass*, *Calcification*, and *Pneumothorax*, may occupy relatively small regions of the image.

We also investigated resizing the images to 380×380 pixels, to the standard input resolution of EfficientNet-B4. However, we considered the additional reduction in spatial resolution to be too costly.

2.2.3 Multilabel target creation.

To perform the classification task, the original annotations were converted into a multi-label classification target. This formulation was chosen because a single chest X-ray can contain multiple abnormalities simultaneously.

Using *one-hot encoding*, a binary indicator was created for each of the 14 classes. The annotation data were then pivoted so that each abnormality became a separate indicator column. To obtain a single target vector for each image, the annotations were aggregated using *logical OR*. This way, if at least one radiologist indicated the presence of a particular abnormality, the corresponding class was assigned a value of 1 for that image.

Initially, the *No Finding* class was excluded from this aggregation. After aggregating the remaining classes, the *No Finding* indicator was assigned a value of 1 only when all other class indicators were equal to 0. However, this approach has an important limitation. If one radiologist reports an abnormality while the other radiologists assign *No Finding*, the image is still labelled as positive for that abnormality. Disagreements between radiologists are resolved in favour of a positive label.

2.2.3 Contrast enhancement and intensity preprocessing.

We investigated three different techniques for modifying the contrast and intensity distribution of the X-ray images: *CLAHE*[2], *percentile-based contrast normalization*, and *gamma correction*. [3]



Figure 5: Visual comparison of a radiograph after applying CLAHE, percentile-based normalization and gamma correction.

These transformations were not applied permanently to the entire dataset. Instead, they were applied on the fly during training. A custom *PyTorch* dataset was implemented in which the selected transformation is applied inside the `__getitem__` method whenever an image is requested for a batch. Consequently, the transformed image is generated when it is loaded rather than being stored as a separate preprocessed image.

This approach avoids creating additional copies of the

dataset and allows different preprocessing strategies to be evaluated without modifying the original images. It's important to state that this preprocessing must be applied to both training images and evaluated images.

2.3 Data Augmentation

Data augmentation is a technique that consists of applying controlled transformations to the training images to artificially increase the variability of the dataset. Its use in this project was motivated by the variability observed in how some radiographs were acquired and processed. These differences, together with the strong class imbalance present in the dataset, could encourage the model to learn image-specific characteristics that are not related to the abnormality being detected.

In our experiments, the *Albumentations*[4] library was used to implement the data augmentation pipeline. We applied transformations such as:

- **Affine transformations:** to introduce small changes in scale and rotation.
- **Contrast:** we tried *CLAHE* and *percentil normalization* techniques, the ones that worked best as preprocessing.
- **Gaussian noise.**

These transformations were applied probabilistically during training, meaning that each image could receive a different combination of transformations in different training iterations. Augmentation was applied only to the training set. Validation images were kept unmodified so that validation performance reflected the model's performance on images closer to the original data distribution.

2.4 Model Architecture.

The classification model is based on EfficientNet-B4 and initialized with weights pretrained on ImageNet. This pretrained parameters are subsequently fine-tuned on the chest X-ray dataset.[5]

The original classification head of EfficientNet-B4 is replaced by a fully connected linear layer. EfficientNet-B4 produces a feature representation of 1,792 features mapped directly to 14 output neurons. The model is formulated as a multi-label classifier and the final layer produces 14 independent logits. During validation, these logits are converted into probabilities using the sigmoid function, with each probability representing the predicted likelihood of the corresponding abnormality.

All model parameters are passed to the optimizer and are therefore updated during backpropagation.

2.5 Training Procedure

The model was trained using the training portion of the dataset and evaluated after each epoch on a held-out validation set. The data was split into 80% training and 20% validation sets using a stratified multilabel split. A batch size of 4 was used due to memory requirements.

The model was trained using *Binary Cross-Entropy with Logits (BCEWithLogitsLoss)* due to the multi-label

formulation of the problema. The loss function combines the sigmoid operation and binary cross-entropy computation in a numerically stable formulation.

The model parameters were optimized using *AdamW optimizer* with:

- Learning rate of $1e-4$
- Weight decay of $1e-4$
- No learning-rate scheduler was used in these experiments.

Macro ROC-AUC and macro PR-AUC were monitored while validation and, both ROC-AUC and PR-AUC were calculated separately for each of the 14 abnormalities. Per-class metrics allow the performance of the model to be evaluated for individual abnormalities, particularly important given the strong class imbalance in the dataset.

Macro PR-AUC was selected as the criterion for model selection because it provides a more informative measure of performance in the presence of imbalanced classes. In addition, given the medical nature of the problem, false negatives are particularly important, as they correspond to abnormalities that are present but not detected by the model. Therefore, achieving a high recall (sensitivity) is a priority. PR-AUC is particularly relevant in this context because it evaluates the trade-off between Precision (Positive Predictive Value) and Recall (True Positive Rate) across different classification thresholds. This allows the model's ability to identify positive cases while controlling the number of false positives to be evaluated without relying on a single arbitrarily selected threshold.

For each experiment, the training and validation loss, macro ROC-AUC, macro PR-AUC, and per-class ROC-AUC and PR-AUC were stored for every epoch.

2.6 Postprocessing

To convert the output probabilities into binary predictions, a threshold must be defined for each class. Rather than using a fixed threshold of 0.5 for all classes, we determined an optimal threshold independently for each abnormality.

The thresholds were selected by searching for the value that maximized the F1-score for each class on the validation set. The F1-score provides a balance between precision and recall and avoids optimizing exclusively for recall, which could result in many false positive predictions.

An alternative threshold-selection strategy was initially investigated. In this approach, a minimum acceptable recall was specified for each class, and the threshold providing the highest precision while satisfying this recall constraint was selected. However, the resulting predictions did not perform as expected.

3 RESULTS

3.1 Evaluation methodology.

The primary metrics used to compare the different experiments were Macro PR-AUC and PR-AUC per class. To further analyse the model's performance, *sensitivity, precision, F1-score, and false positive rate (FPR)* were also considered, with particular emphasis on sensitivity given

the medical nature of the problem. ROC-AUC was retained as a secondary metric.

The objective of the task is to classify each image into one or more of the 14 possible abnormalities. Furthermore, the dataset presents a strong class imbalance. As a result, the number of true negative samples can be considerably larger than the number of positive samples, particularly for the less frequent abnormalities. Metrics such as *accuracy and specificity* can be misleading, as a large number of true negatives can result in high scores even when the model fails to detect a substantial proportion of positive cases. For this reason, PR-AUC and macro PR-AUC were selected as the primary metric.

Macro PR-AUC gives the same weight to each class when computing the overall score, regardless of its prevalence in the dataset. This prevents classes with a large number of instances, such as *Aortic Enlargement*, from dominating the overall evaluation.

3.2 Baseline model.

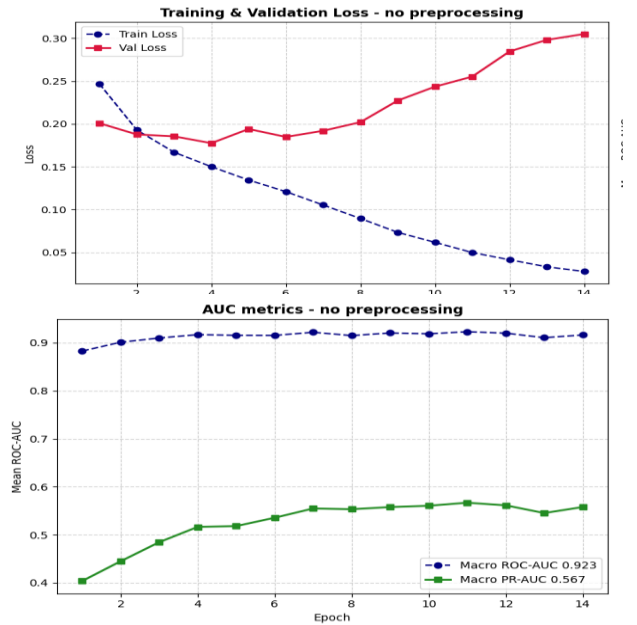


Figure 6: Training Loss value, Validation Loss value, ROC-AUC value and PR-AUC value through every training epoch.

Interestingly, validation loss and macro PR-AUC did not always peak at the same epoch. This repeats throughout all our experiments. Validation loss starts deteriorating around epoch 4, while macro PR-AUC continues improving until epoch 11.

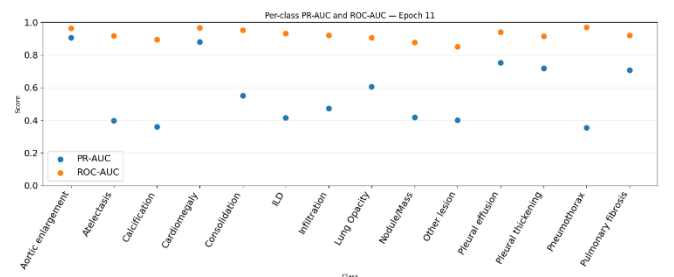


Figure 7: Per class PR-AUC values and per class ROC-AUC values of the baseline

model best epoch (11).

When examining the per-class metrics, we observe that the ROC-AUC scores are relatively high across all classes due to the strong class imbalance (large number of negative samples compared to positive ones). In contrast, the PR-AUC values vary considerably across classes. The classes with higher PR-AUC values generally correspond to those with a larger number of instances in the dataset, while the less frequent classes tend to achieve lower PR-AUC scores.

Other Lesion is an exception to this trend. Despite having a moderate number of instances, it achieves one of the lowest PR-AUC values, at around 0.40. A possible explanation is the high variability within this class. Other Lesion includes lesions that do not belong to any of the other 13 categories, meaning that it represents a heterogeneous group of findings with potentially very different visual characteristics.

3.3 Effect of data augmentation.

Since patient positioning varied across the dataset, we investigated whether introducing small geometric variations during training could improve robustness (rotation and scaling).

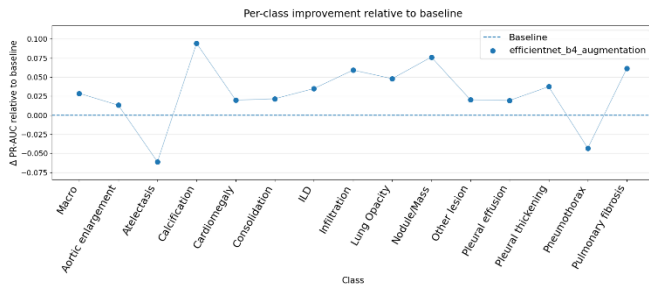


Figure 8: the relative change in PR-AUC for each class with respect to the baseline model.

Interestingly, *Atelectasis* and *Pneumothorax* are the only classes that experienced a decrease in PR-AUC, which coincides with both being among the least frequent classes in the dataset. In contrast, *Pulmonary Fibrosis*, *Calcification*, and *Nodule/Mass* experienced the largest improvements in PR-AUC. *Calcification* and *Nodule/Mass* are abnormalities that may occupy relatively small regions of the image. One possible explanation for their improvement is the introduction of small rotations and scalings during augmentation, which may encourage the model to learn more robust visual characteristics of these abnormalities.

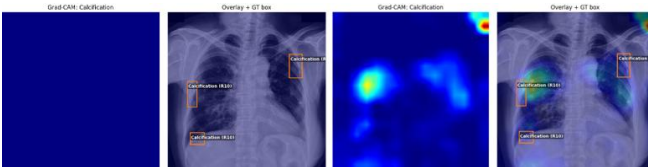


Figure 9: Comparison of Grad-CAM visualizations for the baseline model (left) and the model trained with data augmentation (right). The ground-truth bounding boxes are shown for reference

Calcification was the class that experienced the largest improvement after applying data augmentation. The

Grad-CAM[6] visualizations provide a possible explanation for this improvement. While the baseline model shows very limited activation on the image, the augmented model exhibits stronger activation in regions that partially overlap with the ground-truth calcification areas. However, this activation pattern is not consistently observed across all images.

3.4 Effect of contrast preprocessing

As a next step, different contrast enhancement methods were evaluated to investigate whether modifying the intensity distribution of the radiographs could improve the visual representation of certain abnormalities and consequently help the model learn more discriminative features. The following methods were tested: CLAHE with clip limits of 2 and 3, percentile-based normalization with low and high percentiles of 1 and 99, and gamma correction with gamma values of 1.2 and 0.8.

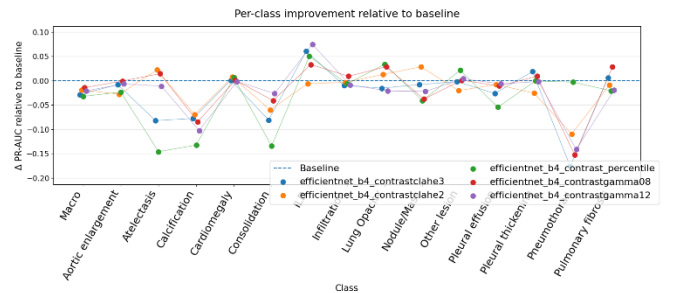


Figure 10: the relative change in PR-AUC for each class with respect to the baseline model.

None of the contrast preprocessing methods provides a consistent improvement across all classes, the effect is highly class-dependent. The largest negative changes are observed for *Atelectasis*, *Calcification*, *Consolidation*, and particularly *Pneumothorax*. This suggests that these contrast transformations may remove information that is relevant for detecting these abnormalities. In the other hand, *Pulmonary Fibrosis*, *Infiltration*, and *Lung Opacity*, show improvements in certain preprocessing methods.

These results indicate that contrast preprocessing does not provide a clear general improvement over the baseline model but certain transformations may be beneficial for specific classes.

3.5 Combining contrast preprocessing and augmentation[7]

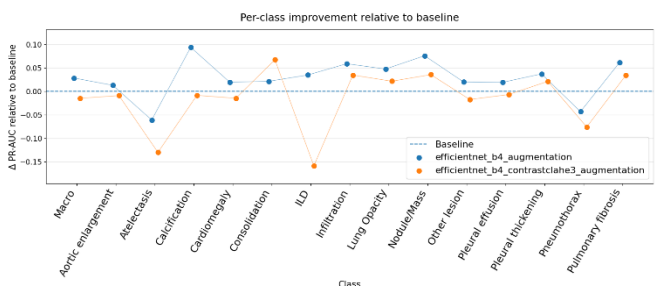


Figure 11: the relative change in PR-AUC for each class with respect to the baseline model

After evaluating the two techniques separately, we investigated whether combining contrast preprocessing with data augmentation based on geometric transformations (rotation and scaling), could provide additional benefits. This combination was motivated by the improvements observed for certain classes when contrast preprocessing was applied independently (ILD).

The combination resulted in a decrease in PR-AUC for every class compared with the augmentation-only approach. Interestingly, *ILD*, which had shown improvement when contrast preprocessing and data augmentation were evaluated separately, experienced one of the largest decreases. This suggests that the effects of the two techniques are not additive and that contrast transformations that may be beneficial when applied independently can interfere with the features learned through geometric augmentation.

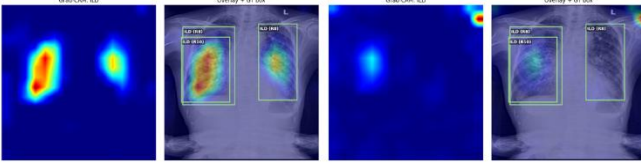


Figure 12: Comparison of Grad-CAM visualizations for the augmentation model (left) and the model trained with data augmentation and contrast preprocessing (right). The ground-truth bounding boxes are shown for reference

3.6 Contrast enhancement as augmentation.

Since data augmentation with geometric transformations produced promising results, we investigated if introducing additional image-level transformations could further improve the model. *Gaussian noise* and *contrast preprocessing* were incorporated into the augmentation pipeline. For contrast preprocessing, we selected the methods and parameters that showed the most promising results when evaluated independently: *percentile-based normalization*, *gamma correction* with $\gamma = 0.8$, and *CLAHE*.

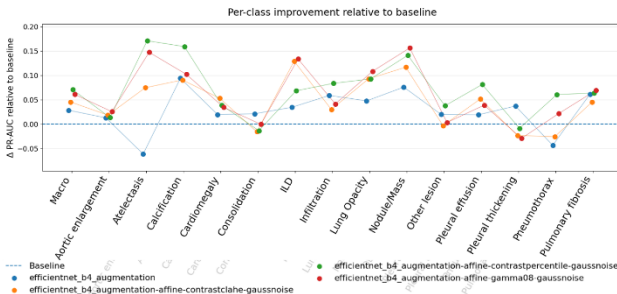


Figure 13: the relative change in PR-AUC for each class with respect to the baseline model.

The results were positive overall, most classes show an improvement in PR-AUC compared with the baseline using only geometric augmentation. The models incorporating *percentile-based normalization* and *gamma correction* ($\gamma = 0.8$), generally achieve larger gains across the different classes.

Overall, these experiments suggest that combining

geometric and intensity-based augmentation is more beneficial than geometric augmentation alone, although the effect depends on the specific preprocessing technique and abnormality.

3.7 Cropping black and white image borders.

Finally, we investigated whether the model's ability to learn relevant abnormality-related features could be improved by removing, as much as possible, the black and white borders surrounding the radiographs. These borders do not contain anatomical information and may introduce image-specific characteristics that are unrelated to the abnormalities being classified.

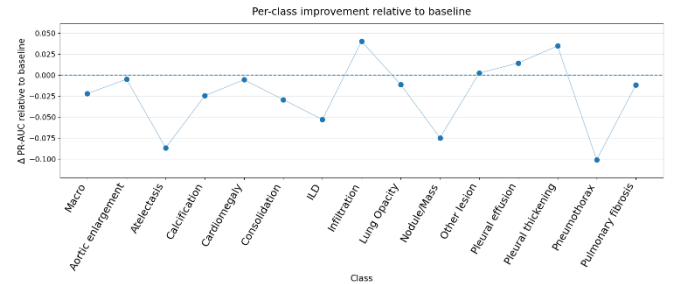


Figure 14: Per-class PR-AUC improvement achieved by the same augmentation pipeline when trained and evaluated on images with cropped borders compared with uncropped images.

The results do not show a consistent improvement compared with the baseline. A possible explanation is that the borders, despite not containing clinically relevant information, may still contain acquisition or preprocessing characteristics that the model has learned to associate with certain classes

Grad-CAM visualizations were also considered to investigate whether removing the borders encouraged the model to focus more frequently on anatomically relevant regions. However, the results did not provide clear evidence that cropping consistently improved the model's attention towards the abnormalities

3.8 Final model.

Our final model: Data augmentation with geometric transformations, percentil-based contrast normalization and gaussian noise.

	roc_auc	pr_auc	precision	recall	f1
Macro	0.935239	0.651009	0.597448	0.650636	0.595971
Aortic enlargement	0.969234	0.918368	0.796226	0.890295	0.840637
Atelectasis	0.940967	0.532529	0.8	0.470588	0.592593
Calcification	0.917535	0.543614	0.586207	0.435897	0.5
Cardiomegaly	0.975662	0.930286	0.701299	0.925714	0.79803
Consolidation	0.935574	0.529399	0.615385	0.484848	0.542373
ILD	0.953955	0.551939	0.517241	0.441176	0.47619
Infiltration	0.954342	0.631646	0.369748	0.830189	0.511628
Lung Opacity	0.929334	0.709582	0.509434	0.716814	0.595588
Nodule/Mass	0.908245	0.51993	0.632653	0.43662	0.516667
Other lesion	0.87129	0.440607	0.303502	0.804124	0.440678
Pleural effusion	0.954583	0.80218	0.788732	0.615385	0.691358
Pleural thickening	0.916567	0.688671	0.574803	0.863905	0.690307
Pneumothorax	0.936765	0.568577	0.416667	0.625	0.5
Pulmonary fibrosis	0.929288	0.746793	0.752381	0.568345	0.647541

Table 1: Final evaluation metrics for the best-performing model.

The final augmentation pipeline achieved an improvement of approximately 0.08 in Macro PR-AUC compared with the baseline model. As shown in Figure X, the improvements were observed across most classes, although the magnitude of the improvement varied considerably depending on the abnormality.

Some classes already achieved strong results with the baseline model, including *Aortic Enlargement*, *Cardiomegaly*, *Lung Opacity*, *Pleural Effusion*, and *Pulmonary Fibrosis*. Although these classes were further improved, the margin for improvement was naturally more limited.

Several of the classes that initially achieved weaker results also experienced the largest improvements, *Atelectasis* and *Nodule/Mass*. This suggests that the proposed augmentations helped the model learn more robust and discriminative features for some of the more challenging abnormalities. In the other hand, *Pneumothorax* and *Other/Lesion* remained challenging to classify. In the case of *Other Lesion*, this may also be explained by the high variability within the class, as it includes lesions that do not belong to any of the other 13 categories.

The results support the relationship observed throughout the experiments between class frequency and model performance but also demonstrate that class frequency is not the only factor affecting performance, as some relatively infrequent classes, benefited substantially from the proposed augmentation strategy.

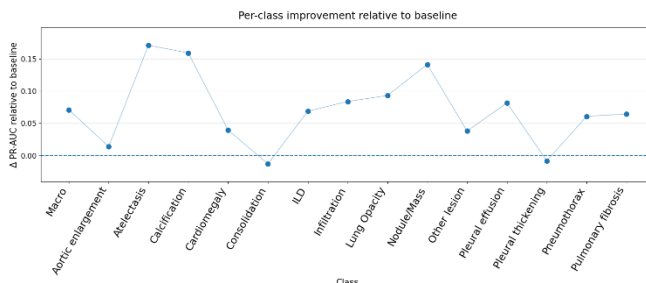


Figure 15: Per-class PR-AUC improvement of the final augmentation pipeline relative to the baseline model.

4 DISCUSSION

4.1 Challenges of the task.

The dataset presented several challenges. Chest X-ray interpretation is difficult, and the annotations were produced by multiple radiologists. Some degree of disagreement or annotation noise may be present in the dataset. In addition, the images showed variability caused by differences in acquisition and preprocessing, which introduced visual characteristics unrelated to the abnormalities of interest. Another important challenge was the strong class imbalance.

Hardware limitations constrained the experiments that could be performed. Training the model was computationally expensive. As a result, the number of architectures and hyperparameter configurations that could be evaluated was limited.

4.2 Comparison with Other Work

A direct comparison with the winning solutions of the original competition is not appropriate, as this project does not address the complete object detection problem.

Nevertheless, reviewing other implementations and exploratory analyses like [Vahe Sokhoyan's](#) dataset would have been valuable during the initial stages of the project. More advanced exploratory data analysis and visualization techniques could have provided additional insights into the distribution of abnormalities, annotation patterns, and characteristics of the images.

4.3 What Could Be Improved in Future Work?

4.3.1 More systematic experiment tracking.

Additional information could have been saved during training and experimentation, including training configurations, hyperparameters, random seeds, learning curves, and per-class metrics. A more systematic experiment tracking process would make comparisons between models easier and improve the reproducibility of the experiments.

4.3.2 Using localization information during training.

Since bounding box annotations are available for the training data, future work could investigate whether localization information can be incorporated into the training process.

4.3.3 Comparing different architectures.

The project focused primarily on EfficientNet as the classification architecture. Future experiments could compare different architectures, such as other convolutional neural networks or transformer-based models, in order to determine whether the observed improvements generalize across architectures.

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